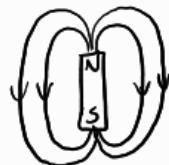


$$\vec{F}_m = e(\vec{v} \times \vec{B}) \Rightarrow \vec{F}_m \perp \vec{v} \Rightarrow A_m = 0$$

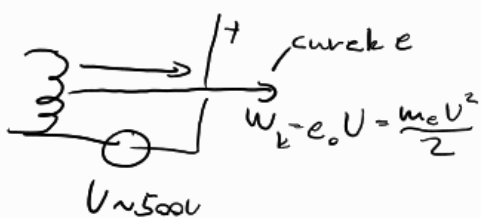
$\vec{F}_m$ je idealna za centripetalne sile
(v hom. magn. polju gibajoč naboj kroži)
če $\vec{v} \nparallel \vec{B}$
- če je prisotna disipativna sila - spirala

$$\vec{v} \Rightarrow \vec{v}_T \perp \vec{B} \Rightarrow F_m = e v_T B = m \frac{v_T^2}{r}$$

$$m v_T = e B r = G_T$$



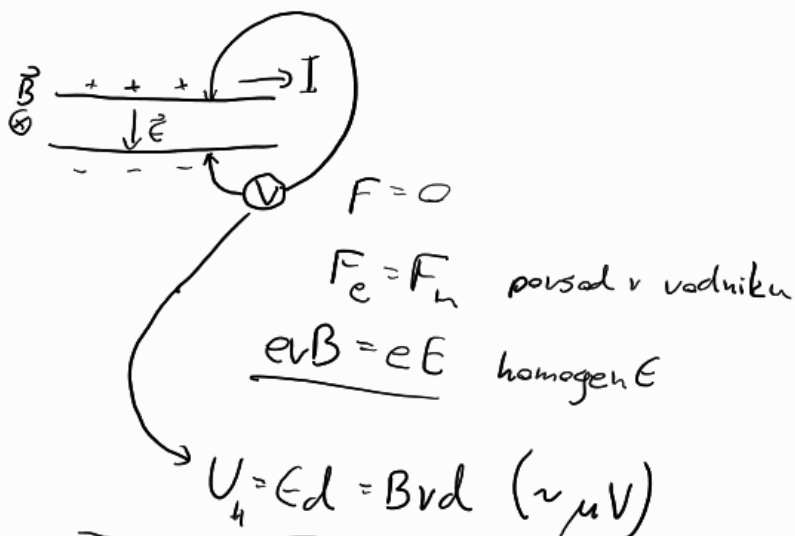
El. top



$$B \left[\frac{Vs}{m^2} = T \right]$$

	$B[T]$
zemlja	$\sim 10^{-4}$
magnet	~ 1
superprevodni magneti	~ 10
neutronska zvezda	10^8

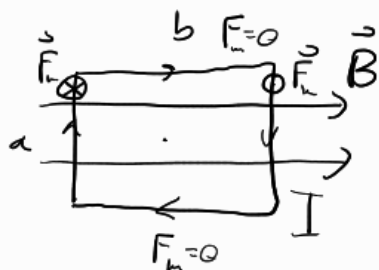
Hall



$$\vec{F}_m = N e_0 \vec{v} \times \vec{B} = e_0 n_e S l \vec{v} \times \vec{B}$$

$$\vec{F}_m = \vec{I} \times \vec{B}$$

$$I = \frac{de}{dt} = e_0 \frac{dN}{dt} = e_0 n_e S v$$



$$\vec{M} = (Fb = IaBb) = \vec{I} \vec{S} \times \vec{B} = \vec{p}_m \times \vec{B}$$

$$W_m = -\vec{p}_m \cdot \vec{B}$$

↑
magnetni dipol

$$p_m [Am^2 = \frac{J}{T}]$$

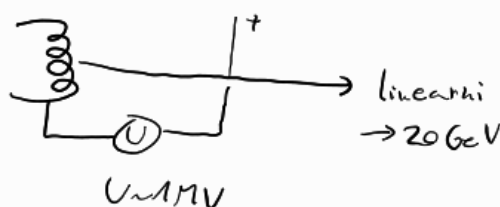
$$p_e [Asm]$$

$$\vec{M} = \vec{p}_e \times \vec{E}$$

↓
dipol

$$W_e = -\vec{p}_e \cdot \vec{E}$$

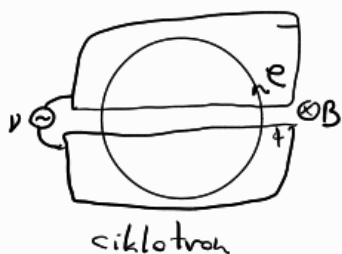
Pospeševalnik: $W_k = eV$



$$mv = eBr$$

$$\nu = \frac{1}{t_0} = \frac{2\pi r}{v}$$

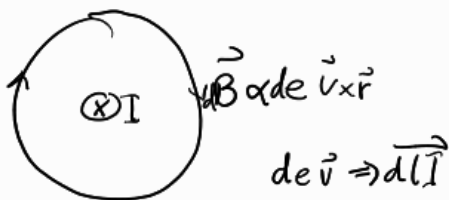
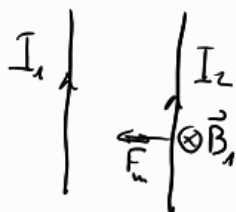
$$= 2\pi \frac{eB}{m}$$



(treba je 2x zamenjati + in - ko e naredi en krog)

$$\nu = 2\pi \frac{e_0 \hbar}{m_e} \approx 10^{12} Hz \quad \text{Preveč!}$$

Za protone: $\nu = 500 MHz \checkmark \quad (\sim 10 MV)$



$$d\vec{B} \propto \frac{d\vec{l} \hat{I} \times \vec{r}}{r^3}$$

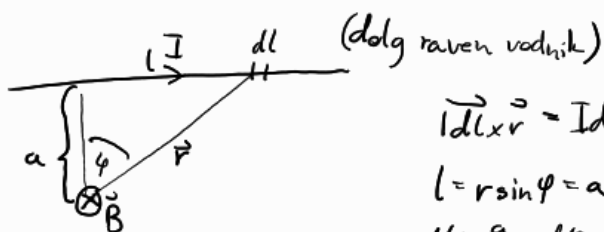
$$\left[\frac{Vs}{m^2} \dots \frac{Am}{m^2} \right]$$

$$\boxed{dB = \frac{\mu_0 d\vec{l} \hat{I} \times \vec{r}}{4\pi r^3}}$$

Biot-Savartov zakon

$$\mu_0 \left[\frac{Vs}{Am} \right] = 4\pi \cdot 10^{-7} \frac{Vs}{Am}$$

magn. konst.
(indukcijska konst.)



(dolga raven vodnik)

$$|d\vec{l} \times \vec{r}| = |d\vec{l}| r \cos \varphi = I \frac{a}{\cos^2 \varphi} d\varphi \frac{a}{\cos^2 \varphi} \cos \varphi$$

$$l = r \sin \varphi = a \tan \varphi$$

$$dl = \frac{a}{\cos^2 \varphi} d\varphi$$

$$r = \frac{a}{\cos \varphi}$$

$$dB = \frac{\mu_0 I}{4\pi} \frac{dl}{r^2} \frac{\cos \varphi}{a} d\varphi = \frac{I}{a} \cos \varphi d\varphi$$

$$B = \frac{\mu_0 I}{2\pi a}$$

Ampereov zakon

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I$$

(Integral po zaklj. poti)

Gaussov zakon za mag. polje

$$\oint \vec{B} \cdot d\vec{s} = 0$$

Tuljava

$$B = \frac{\mu_0 N I}{2\pi r}$$

$$\vec{v}_1 \uparrow \quad \vec{v}_2 \uparrow$$

$$\vec{e}_1$$

$$\vec{e}_2$$

$$\vec{F}_{m_{2 \rightarrow 1}} = \vec{e}_2 \vec{v}_2 \times \vec{B}_1 = \vec{e}_2 \vec{v}_2 \times \frac{\mu_0}{4\pi} \frac{e_1}{r^3} (\vec{v}_2 \times \vec{e}_1)$$

$$\vec{F}_e = \frac{e_1 e_2}{4\pi \epsilon_0 r^2} \frac{\vec{r}}{r}$$

$$\frac{1}{\epsilon_0} \leftrightarrow \mu_0 v^2 \rightarrow \boxed{\frac{1}{\epsilon_0 \mu_0} = c^2}$$

$$\frac{F_m}{F_e} = \left(\frac{v}{c}\right)^2 \rightarrow F_m \ll F_e \text{ za gibanje dveh nabojev}$$

$$\vec{F}_{m_{2 \rightarrow 1}} = \vec{e}_1 \vec{v}_1 \times \frac{\mu_0}{4\pi} \frac{e_2}{r^3} (\vec{v}_2 \times \vec{e}_1)$$